Get Input Matrices From an Image Vector y.

Define a transformation $T : \mathbf{R}^{n\times n} \to \mathbf{R}^{n(n+1)/2}$ given by

$$y = T(X), \quad \text{where } Z = \frac{(X + X^T)}{2},$$
$$y = (z_{11}, z_{12}, \ldots, z_{1n}, z_{22}, z_{23}, \ldots, z_{2n}, z_{33}, \ldots, z_{nn}, z_{n-1,n-1}, z_{n-1,n}, z_{nn}).$$

For any vector $y \in \mathbf{R}^{n(n+1)/2}$, is it possible to find N *distinct* matrices X that is mapped to y? Explain your concept in details (in the HW paper).

If the answer is possible, write a function `getXfromy(y, N, n)` that returns N distinct matrices X that $y = T(X)$. Your function should work for arbitrary n, N and y.

Input Format

- N: a positive integer where $10 \leq N \leq 50$
- $y \in \mathbf{R}^{n(n+1)/2}$ where $5 \leq n \leq 100$

Output Format

- $X \in \mathbf{R}^{n\times n\times N}$: The N matrices of size $n\times n$. You must ensure that $T(X[:,:,k]) = y$ for all $k = 1, \ldots, N$. Moreover, these N matrices are all distinct.

Implementation

Goal: Fill in the following function:

```
import numpy as np

def getXfromy(y, numX, n):
    ...
    return allX
```

Hint

- Answer to the conceptual questions in HW paper first. This question is not about heuristically generating X. Your solution relies on the linear transformation concept in the class.